

Security Incident Report

Section 1: Identify the network protocol involved in the incident

Receiving IP requests SYN package request sent to DNS server.
DNS server responds with SYN-ACK package request with correct IP address.
DNS server also responds with malware package.
Receiving IP initiates download of malware and sends ACK package request to gain access to HTTP address of greatreceipesforme.com

Section 2: Document the incident

A brute-force attack was executed to gain access to the administrative account for yummyrecipesforme.com
Customers reported that when attempting to gain access to the DNS domain that a prompt for a downloadable file for access was prompted. After selecting the file, customers reported that the website changed and their personal computers began to run slowly.
The website owner attempted to access the admin panel but was unable to.
Security teams created a sandbox environment and used the network protocol analyzer tcpdump to observe the issue. In the tcpdump logs, records indicate initial requests from receivers IP request to the DNS server were correct, however, in the SYN-ACK package, receivers received a malware package. In the ACK package, the DNS server initiates responds with an HTTP address to a fake website greatreciepesforme.com
Upon investigation, it is concluded the website has been compromised. Upon looking at the source code, it is found that added javascript code was embedded into the website in an executable file, redirecting customer traffic away from DNS server to the malware DNS file.

Section 3: Recommend one remediation for brute force attacks

Recommendation is to temporarily disable the website to allow teams to remove the javascript source code from the admin permissions.

After removing the javascript, it is recommended to update the password policies of the company to ensure passwords are routinely changed, disallowing previous passwords from being used, and limiting the number of login attempts to have a system in place to prevent against brute force attacks.